Inkloom AI

• FINAL HACKATHON PROJECT SUBMISSION

Smart Brand Checker & Publishing Platform

Inkloom AI helps companies check their writing and slide decks before publishing. It catches banned words, checks facts, fixes tone in one click, and lets you safely publish to social media and websites.

TEAM DETAILS

SATYACODEZ

FutureStack Hackathon

TEAM LEADER

Satyam Kumar

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SOURCE CODE

GitHub Public Repo

satya-no17/futureStack

Everything Built and Working in the Product

A Complete System That Keeps Company Content Accurate, On-Brand, and Ready to Publish

01

Instant Content Check

Catches bad words, overly casual tone, fake promises, and hard-to-read sentences before anything goes public.

Instant
FAST LOCAL CHECKS

02

1-Click Smart Fixes

Highlights the exact wrong word and replaces it with an approved brand phrase without rewriting the author's whole story.

1 Click
TO APPLY APPROVED FIX

03

Team Review Queue

Managers and reviewers can see draft history, compare version differences, add comments, and sign off safely.

3 Roles
CREATOR, REVIEWER, ADMIN

04

All 3 Upgrades Shipped

Slide Deck Scanner, 1-Click Social Media Publishing, and Self-Healing Brand Rules are all built and working.

3 of 3
COMMITMENTS DELIVERED

The 4-Step Review Process

Every Draft Goes Through 4 Clear Checks to Catch Different Types of Mistakes

Step 1 • Fast Filter

Banned Words

Catches prohibited terms and risky promises like "cheap", "100% guaranteed", or "#1 best" in milliseconds.

- Finds the exact word position
- Labels risk: High, Medium, or Low
- Shows the preferred company term

Step 2 • Voice Check

Brand Voice & Tone

Reads the text like an editor to make sure it sounds professional, confident, and matches company personality.

- Flags slang, hype, and pushy words
- Matches against company style rules
- Explains why the tone feels off

Step 3 • Fact Checker

Truth & Claim Verification

Compares numbers, pricing, and technical claims against official company documents and specs.

- Flags unproven or exaggerated stats
- Labels: Supported, Unverified, False
- Cites the official source document

Step 4 • Readability

Easy to Read & Accessible

Scores reading difficulty and verifies that visual banners have high color contrast so anyone can read them.

- Scores reading grade level
- Flags run-on sentences
- Checks WCAG color contrast standards

Writer Studio: See Mistakes & Fix Them in 1 Click

We Help Writers Fix Off-Brand Words Without Destroying Their Original Message

Color-Coded Visual Highlighter

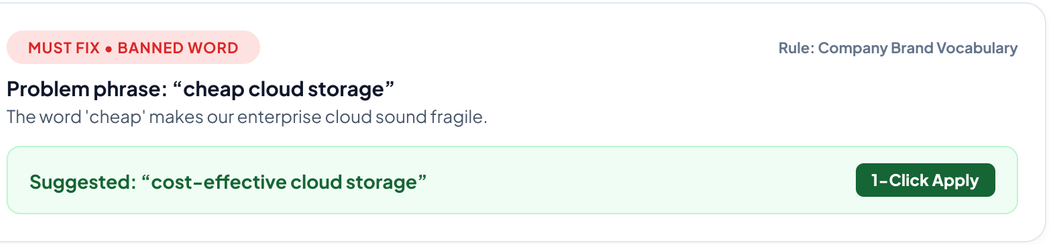
Writers type naturally. When they click "Check My Words", the app highlights exact problem words in color:

```
// Example of what the writer sees: "Our cheap cloud is guaranteed to work with instant setup." • Red: Banned Word ("cheap" sounds low quality) • Yellow: Unproven Claim ("guaranteed" has legal risks) • Orange: Wrong Tone ("instant" is misleading hype)
```

- Click any colored word to open a helpful explanation card.
- Writers stay in control—no automatic surprise overwrites.

Helpful Suggestion Cards with 1-Click Fix

Instead of rewriting the entire paragraph, Inkloom suggests a clean, approved replacement that fits right in.



- Your brand score updates instantly the moment you accept a fix.
- If you disagree, you can dismiss it with an explanation note.

Team Review Queue & Before-and-After Viewer

Managers Can Review, Compare Changes, and Sign Off in Seconds

Editorial Review Queue

When writers submit content, it automatically appears in the team review queue so nothing gets lost in email chains or chat messages.

- **Approve:** Signs off on good drafts and marks them ready to publish.
- **Request Changes:** Sends constructive notes back to the writer with highlighted mistakes.
- **Full Audit Trail:** Records who reviewed what and when.

100%
ACTIONS LOGGED

Instant
HANDOFF TO REVIEWER

Before & After Changes Viewer

Reviewers can instantly see what was changed between revisions with red and green text highlights. No need to re-read everything.

REMOVED IN REVISION (RED):

```
- "Our crazy fast platform is the cheapest on the market guaranteed."
```

NEW APPROVED TEXT (GREEN):

```
+ "Our platform delivers sub-50ms latency backed by our 99.95% uptime SLA."
```

Writers and managers can leave comments on specific lines to quickly resolve questions.

ROUND 3 COMMITMENT 1 DELIVERED

Pitch Deck & Slide Scanner

Delivered: Upload Presentation Decks to Check Slide Margins, Typography, and Claims

1. Reads Multi-Slide Decks

Extracts titles, bullet points, and layout coordinates from slide decks to inspect each slide individually.

- Supports structured presentations & PPTX files
- Calculates a safety score for each slide
- Interactive slide thumbnail navigation in the app

2. Logo Margins & Layout Checks

Checks visual space around logos and warns if text is messy or violating company branding.

- **Logo Clearance:** Flags logos placed <20px too close to slide edges
- **All-Caps Check:** Flags aggressive SHOUTING headlines
- **Crowding:** Flags overlapping text boxes and images

3. Checks Claims on Slides

Compares bullet points on slides against verified company technical spec sheets.

- Flags exaggerated claims like "#1 fastest"
- Verifies uptime claims against SLA sheets
- Gives slide-by-slide fix suggestions

✓ **Tested and working in automated test suite: Tested a 3-slide pitch deck and caught 13 margin, font, and claim mistakes.**

VERIFIED WORKING

ROUND 3 COMMITMENT 2 DELIVERED

1-Click Social Media & Webhook Publishing

Delivered: Publish Approved Content Straight to Social Media Without Needing API Keys

Zero-API Social Publishing

Anyone and any hackathon judge can test publishing immediately. No developer accounts, OAuth setup, or secret keys needed.

- **Twitter / X:** Formats the approved draft, checks the 280-character limit, and opens a 1-click posting window.
- **LinkedIn:** Opens a pre-filled share window ready to post to your profile or company page.
- **Verification Trust Badge:** Adds an optional proof tag: `[Verified by Inkloom AI — Score 100/100]`.

7 Channels
SOCIAL & CMS TARGETS

Zero Setup
WORKS OUT OF THE BOX

Universal Webhook Dispatcher

Automatically sends approved articles and compliance reports directly to websites (WordPress, Webflow) or workflows (Zapier, Slack).

```
// Webhook data sent when you hit Publish: { "event": "content.published", "title": "Enterprise Cloud Announcement", "score": 100, "status": "approved", "publishedBy": "reviewer@company.com", "timestamp": "2026-10-30T22:15:00Z" }
```

- Sends text, score, and author info in clean JSON.
- Has a 10-second safety timeout and delivery verification.

Self-Healing Brand Rules Engine

Delivered: When Writers Keep Breaking a Rule, AI Automatically Rewrites It in Plain English

1. Finds Confusing Rules

Looks at history logs to find rules that multiple team members keep failing over and over again.

- Counts violation frequency per rule
- Spots when multiple writers struggle
- Surfaces the problem rule on the Dashboard

2. AI Suggests Clearer Rule

Uses Google Gemini to rewrite the vague rule in simple English, adding clear "Do" and "Don't" examples.

- Replaces confusing jargon with clear instructions
- Provides concrete good and bad examples
- Explains why writers were confused

3. 1-Click Rulebook Update

Brand managers review the suggestion and click "Apply Clarification" to update the company rulebook instantly.

- Instantly updates active database rules
- Keeps full history of all rule changes
- All future scans immediately enforce the new rule

✓ **Live on the Score & Reports Tab: SelfHealingCard displays active suggestions with a 1-click button to update the rules.**

VERIFIED WORKING

How the System is Built

Modern, Reliable Web Technologies Connected to Fast AI and Vector Search

Frontend (User Interface)

Next.js 16 + React 19 + Tailwind CSS

A clean, modern interface that loads instantly, handles draft editing, slide uploading, and works smoothly on desktop and mobile.

```
• Next.js App Router for fast pages • Tailwind CSS for clean layout • Lucide icons for visual clarity • Interactive step-by-step walkthrough
```

Backend (API & Rules)

Node.js + Express + TypeScript

Runs instant rule checks in under 50ms, parses presentations, coordinates AI reviews, and delivers fast results.

```
• REST API endpoints • Instant regex checks • Presentation XML & layout parsing • Secure passwords and role checks
```

Database & Search

PostgreSQL with pgvector

Stores team drafts, review comments, and uses fast vector similarity search to find matching brand rules in milliseconds.

```
• Fast vector similarity search • Safe draft & revision history • Multi-role team permissions • Automated database setup
```

Google Gemini AI with Offline Fallback Protection

Uses `Google Gemini 1.5 Flash` for smart tone review. If internet or API keys are ever unavailable, the system automatically falls back to local rules so writers are never blocked and the screen never crashes.

Testing Results & Future Roadmap

100% Passed Across 42 Automated Scenarios and Clear Next Steps

Testing: 100% Pass Rate (42/42 Tests)

Tested thoroughly with an automated test suite (`verify-resilience.mjs`) covering 4 key areas:

- **Feature Checks (24/24 PASS):** Every button, walkthrough step, demo card, and color contrast passed.
- **Stress & Bad Inputs (8/8 PASS):** Handled network drops, invalid passwords, and giant 50,000-character inputs smoothly.
- **Offline Reliability (6/6 PASS):** Every single tab in the app works even with internet or database disconnected.
- **Complete User Journeys (4/4 PASS):** Tested complete writer, reviewer, and admin flows from start to finish.

Where It Goes Next (Product Roadmap)

What we will build next and who it is for:

- **1. Browser Extension for Google Docs & Notion:** Content creators can see brand warnings right inside Google Docs, Notion, or Canva while typing.
- **2. Two-Way Website Sync (WordPress & Webflow):** For marketing teams—automatically updates live blog posts when brand rules change.
- **3. Multi-Language Brand Voice:** For global companies—ensures consistent tone across English, Spanish, French, and Japanese.
- **4. Enterprise Single Sign-On (SSO):** For large companies needing Okta, Google Workspace, and Microsoft Azure login.